

Graph of a Two-Variable Function: Definition

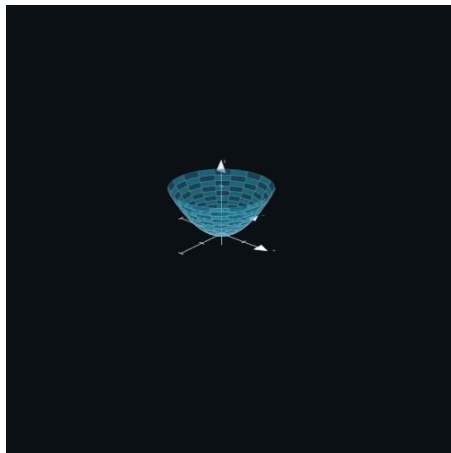
Definition (Graph in $\mathbb{R}^3$)

The **graph** of a function of two variables f with domain D is the set of all points $(x, y, z) \in \mathbb{R}^3$ such that $z = f(x, y)$ and $(x, y) \in D$:

$$\text{Graph}(f) = \{(x, y, z) \in \mathbb{R}^3 \mid z = f(x, y), (x, y) \in D\}.$$

Geometric Transition ($2D \rightarrow 3D$)

- In single-variable calculus, the graph of $y = f(x)$ is a **curve** in $\mathbb{R}^2$.
- In multivariable calculus, the graph of $z = f(x, y)$ is a **surface** S in $\mathbb{R}^3$.



From 2D curves to 3D surfaces

The Vertical Line Test in $\mathbb{R}^3$

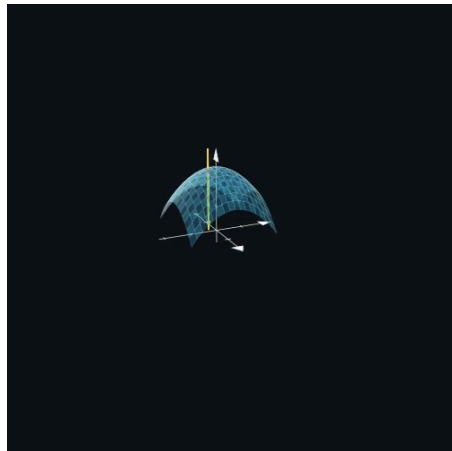
The Vertical Line Criterion

A surface $S \subset \mathbb{R}^3$ represents the graph of a single-valued function $z = f(x, y)$ if and only if:

- Every **vertical line** through $(x_0, y_0) \in D$ (parallel to the z -axis) intersects the surface S at **exactly one point**:

$$P = (x_0, y_0, f(x_0, y_0)).$$

- If a vertical line intersects the surface at more than one point, the surface **cannot** be expressed as a single function $z = f(x, y)$.



Single intersection (x_0, y_0, z_0) per vertical line

Fundamental Surface Families

Standard Quadric & Explicit Surfaces

- **Plane:** $z = ax + by + c$
- **Circular Paraboloid:** $z = x^2 + y^2$
- **Elliptic Paraboloid:** $z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$
- **Circular Cone:** $z = \sqrt{x^2 + y^2}$
- **Upper Hemisphere:** $z = \sqrt{R^2 - x^2 - y^2}$
- **Hyperbolic Paraboloid (Saddle):**

$$z = x^2 - y^2$$



Hyperbolic Paraboloid (Saddle Surface)